

Music Era and Release-Market Classification

DeepMIR 2026 Fall — Homework 1

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Code, checkpoints, README: <https://github.com/chijenpeng/26-fall-DeepMIR-HW1>

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Experimental Setting

Pipeline

- 30 s clip, 24 kHz mono
- frozen encoder → mean + std pooling over time
- StandardScaler → L2 logistic regression
- train = fit, validation = model selection, test = prediction file only

Hardware / software

GPU	2 × NVIDIA RTX A4500 (20 GB)
CPU / RAM	Intel i9-14900K, 125 GB
OS	Ubuntu 26.04
Software	PyTorch 2.14, Transformers 4.45, scikit-learn 1.9

Results

Task 1: release decade

Model: frozen encoders + logistic regression

- Whisper-tiny encoder, **layer 1**
- MERT-v1-330M, **layers 5, 6, 14**
- mean + std pooling over time (6,912-d)

Validation, 132 clips	Top-1	Top-3
Best model	52.3%	86.4%

Errors

- **57%** of errors land on a **neighboring decade**
- Hardest: 2000s (recall .23), absorbed by 2010s and 1990s
- Easiest: 1960s and 2010s, which have one neighbor each

True	1960s	17	2	0	3	0	0
	1970s	5	9	3	2	2	1
	1980s	2	2	13	2	3	0
	1990s	1	3	2	9	4	3
	2000s	2	2	1	5	5	7
	2010s	0	2	0	0	4	16
		1960s	1970s	1980s	1990s	2000s	2010s
		Predicted					

Counts. Rows = true decade (22 clips each).

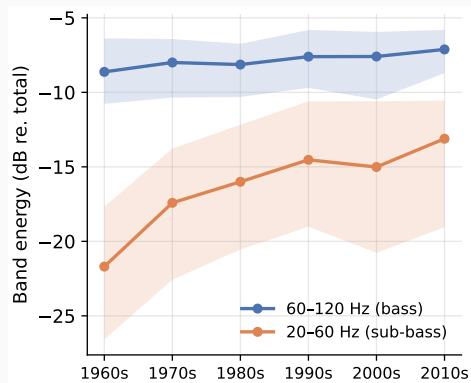
Why neighbors get confused (1): the low end

Hypothesis. Old masters were cut for vinyl, where deep bass makes the stylus jump, so they should carry less sub-bass.

Analysis. Long-term average spectrum (Welch) of the 1,026 training clips, split into two bands.

dB re. total	60s	70s	80s	90s	00s	10s
20-60 Hz	-21.7	-17.4	-16.0	-14.5	-15.0	-13.1
60-120 Hz	-8.6	-8.0	-8.1	-7.6	-7.6	-7.1

- Sub-bass rises by 8.6 dB from the 1960s to the 2010s; the 60-120 Hz band barely moves
- From the 1990s on the low end is nearly identical, so it cannot separate 1990s / 2000s / 2010s



Median and inter-quartile range, training split.

Why neighbors get confused (2): the loudness war

Hypothesis. From the 2000s, brick-wall limiters push masters louder and flatten the peaks. **Analysis.** Waveform statistics of the training clips.

Median per decade	60s	70s	80s	90s	00s	10s
RMS level (dBFS)	-14.0	-14.6	-14.5	-13.0	-11.0	-10.6
Crest factor, peak – RMS (dB)	13.9	14.5	14.5	13.0	11.0	10.6
Dynamic range, loudest 20% blocks (dB)	10.6	11.2	11.3	10.7	8.9	8.7
Longest run pinned at the peak (samples)	5	5	6	9	22	26
Samples within 0.5 dB of the peak (%)	0.02	0.01	0.02	0.06	0.42	0.50

- 2000s and 2010s masters are about 3 dB louder with 3 dB less crest, and show the limiter signature: peaks held at the ceiling
- The two decades are **almost indistinguishable** on every dynamics measure, matching the 2000s ↔ 2010s confusion
- Production cues only split the data into “old” and “new”; finer distinctions must come from musical content

Task 2: release market

Model: frozen encoders + logistic regression

- Whisper-large-v3 encoder, **layers 23, 30**
- MERT-v1-95M, **layer 7**
- Whisper-small language-ID posteriors

Validation, 102 clips	Top-1	Top-3
Best model	60.8%	90.2%

Errors

- 75%** of errors are among **US, UK, Germany, Italy**
- Largest confused pair: US $\leftrightarrow$ UK (9 clips)
- Brazil (.88) and Spain (.82) are rarely missed
- Cause: English-language releases dominate Germany (83%) and Italy (55%)

True	US	11	4	0	0	1	1
	UK	5	9	0	0	2	1
	Brazil	0	0	15	0	0	2
	Spain	0	2	0	14	1	0
	Germany	3	4	1	2	6	1
	Italy	1	4	0	2	3	7
		US	UK	Brazil	Spain	Germany	Italy
		Predicted					

Counts. Rows = true market (17 clips each).

Optional Experiments

Input length: 5 / 10 / 15 / 30 s

Ablation

Frozen features vs. fine-tuning

Audio Language Model

Disclosure

References
